

Multiple Choice

1. **Answer:** C

Options: A) $101.1 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$; B) $26.75 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$; C) $7.081 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$; D) $0.9377 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$

Note: Correct option: C - $7.081 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$

2. **Answer:** D

Options: A) T_1 , unlike T_2 , is sensitive to very low-frequency molecular motions.; B) T_2 , unlike T_1 , is sensitive to very low-frequency molecular motions.; C) T_1 , unlike T_2 , is sensitive to molecular motions at the Larmor frequency.; D) T_2 , unlike T_1 , is sensitive to molecular motions at the Larmor frequency.

Note: Correct option: D - T_2 , unlike T_1 , is sensitive to molecular motions at the Larmor frequency.

3. **Answer:** D

Options: A) I only; B) II only; C) III only; D) I and III only

Note: Correct option: D - I and III only

4. **Answer:** A

Options: A) Neutrons; B) Alpha particles; C) Beta particles; D) X-rays

Note: Correct option: A - Neutrons

5. **Answer:** C

Options: A) Nuclear magnetic resonance; B) Infrared; C) Raman; D) Ultraviolet-visible

Note: Correct option: C - Raman

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It is a red-brown volatile liquid.'

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'Cu'.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Tl. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** a reducing agent. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key